

Sodium Silicate Solution, MR 2.6-3.2

ACCORDING TO WORK HEALTH AND SAFETY (WHS) REGULATIONS

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

Product Name Sodium Silicate Solution, MR 2.6-3.2
 Synonym(s) Silicic Acid, Sodium Salt Solution; Water Glass; Soluble Glass.
 N-Grade, O-Grade, JHG-Grades, NG-Grade
 CAS No. Not applicable.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified Use(s) Detergent ingredient, Adhesive, Binder, Feedstock silica source, General chemical.
 Uses Advised Against Not known.

1.3 Details of the supplier of the safety data sheet

Manufacturer
 Head Office Potters Industries Pty Ltd
 Address 9-13 Rhur Street
 PO Box 4380
 Dandenong South
 VICTORIA 3164
 Telephone (03) 9708 9200
 Fax (03) 9708 9250
 E-mail sds@pottersindustries.com
 Branch N.S.W. Branch
 Address 8-10 Riverside Road
 Chipping Norton
 New South Wales 2170
 Telephone (03) 9708 9200
 E-mail sds@pottersindustries.com

1.4 Emergency telephone number

Chemtrec (Australia) +61 2 9037 2994
 Chemtrec (New Zealand) +(64)-98010034

SECTION 2: HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

In accordance to GHS as implemented under WHS Regulations Skin Irrit. 2 :Causes skin irritation.
 Eye Irrit. 2A :Causes serious eye irritation.

2.2 Label elements

In accordance to GHS as implemented under WHS Regulations

Product Name Sodium Silicate Solution, MR 2.6-3.2
 Hazard Pictogram(s)



GHS07

Signal Word(s) Warning
 Hazard Statement(s) H315: Causes skin irritation.
 H319: Causes serious eye irritation.

Precautionary Statement(s) P264: Wash hands and exposed skin thoroughly after handling.
 P280: Wear protective gloves/protective clothing/eye protection/face protection.

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P302+P352: IF ON SKIN: Wash with plenty of water.
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes.
Remove contact lenses, if present and easy to do. Continue rinsing.
P332+P313: If skin irritation occurs: Get medical advice/attention.
P337+P313: If eye irritation persists: Get medical advice/attention.

2.3 Other hazards

None known.

2.4 Additional Information

For full text of H/P Statements see section 16.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**3.1 Substances**

Not applicable.

3.2 Mixtures

HAZARDOUS INGREDIENT(S)	CAS No.	%W/W	Hazard Statement(s)	Hazard Pictogram(s)
Silicic acid, sodium salt (MR 2.6-3.2)	1344-09-8	30 - 60	Skin Irrit. 2 H315 Eye Irrit. 2A H319	GHS07

For full text of H/P Statements see section 16.

SECTION 4: FIRST AID MEASURES**4.1 Description of first aid measures**

Inhalation If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing.

Skin Contact Wash with plenty of water. Take off contaminated clothing and wash it before reuse.
If skin irritation occurs: Get medical advice/attention.

Eye Contact Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.

Ingestion Rinse mouth. Give plenty of water to drink. Do not induce vomiting. Seek medical advice.

4.2 Most important symptoms and effects, both acute and delayed

May cause irritation.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES**5.1 Extinguishing media**

Suitable Extinguishing media Extinguish with carbon dioxide, dry chemical, foam or waterspray.
Unsuitable extinguishing media None.

5.2 Special hazards arising from the substance or mixture

Non-flammable. Flammable hydrogen gas may be produced on prolonged contact with metals such as aluminium, tin, lead, and zinc.

5.3 Advice for firefighters

Fire fighters should wear complete protective clothing including self-contained breathing apparatus.

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SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Provide adequate ventilation. Stop leak if safe to do so. Avoid contact with skin and eyes. Caution - spillages may be slippery. Wear appropriate personal protective equipment, avoid direct contact.

6.2 Environmental precautions

Do not allow to enter drains, sewers or watercourses.

6.3 Methods and material for containment and cleaning up

Adsorb spillages onto sand, earth or any suitable adsorbent material. Transfer to a container for disposal. Cautiously neutralize remainder. Then wash away with plenty of water.

6.4 Reference to other sections

See Also Section 8, 13.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Avoid contact with skin and eyes. Wash hands and exposed skin thoroughly after handling. Wear protective gloves/protective clothing/eye protection/face protection. Do not eat, drink or smoke when using this product.

7.2 Conditions for safe storage, including any incompatibilities

Keep container tightly closed.

Storage temperature

5 - 60°C

Loading temperature

5 - 50°C

Storage life

12 Month(s). (Minimum)

Incompatible materials

Do not store in aluminum, fiberglass, copper, brass, zinc or galvanized containers. Store in clean steel or plastic containers. Separate from acids, reactive metals, and ammonium salts.

7.3 Specific end use(s)

Detergent ingredient, Adhesive, Binder, Feedstock silica source, General chemical.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

8.1.1 Occupational Exposure Limits

SUBSTANCE.	CAS No.	LTEL (8 hr TWA ppm)	LTEL (8 hr TWA mg/m³)	STEL (ppm)	STEL (mg/m³)	Note
						None assigned

Source: Workplace Exposure Standards For Airborne Contaminants, 2019, Australia

8.2 Exposure controls

8.2.1. Appropriate engineering controls Ensure adequate ventilation. A washing facility/water for eye and skin cleaning purposes should be present.

8.2.2. Personal protection equipment



Eye Protection

Wear goggles giving complete eye protection. (AS/NZS 1337.1)

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Skin protection

Wear protective gloves. (ANS/NZS 2161.10)

Breakthrough time of the glove material: refer to the information provided by the gloves' producer.



Respiratory protection

Normally no personal respiratory protection is necessary. In case of inadequate ventilation wear respiratory protection.



Thermal hazards

Take appropriate precautions when handling bulk product that is transported/shipped whilst hot as it can cause thermal burns.

Wear insulating gloves (ANS/NZS 2161.4)

8.2.3. Environmental Exposure Controls Do not allow to enter drains, sewers or watercourses.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Physical state	Liquid.
Colour	Clear - hazy - Colourless.
Odour	Odourless.
Melting point/freezing point	0°C
Boiling point or initial boiling point and boiling range	105-108°C
Flammability	Not known.
Lower and upper explosion limit	Not known.
Flash Point	Not known.
Auto-ignition temperature	Not known.
Decomposition Temperature	Not known.
pH	11 - 13 (100% Sodium Silicate)
Kinematic Viscosity	20 - 5000 cPs
Solubility	Solubility (Water) : Soluble in water. Solubility (Other) : Not known.
Partition coefficient n-octanol/water (log value)	Not known.
Vapour pressure	Not known.
Density and/or relative density	Relative density : 1.2 - 1.6
Relative vapour density	Not applicable.
Particle characteristics	Not known.

9.2 Other information

Percent Volatile (%):	30 - 60%
Volatile Organic Compound Content (%):	0

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

May react violently with: Acids.

10.2 Chemical Stability

Stable under normal conditions. Absorbs carbon dioxide on exposure to air, which results in the deposition of insoluble silica.

10.3 Possibility of hazardous reactions

No hazardous reactions known if used for its intended purpose.

**Sodium Silicate Solution, MR 2.6-3.2****10.4 Conditions to avoid**

Carbon dioxide.

10.5 Incompatible materials

Do not store in aluminum, fiberglass, copper, brass, zinc or galvanized containers.
Separate from acids, reactive metals, and ammonium salts.

10.6 Hazardous decomposition products

Flammable hydrogen gas may be produced on prolonged contact with metals such as aluminium, tin, lead, and zinc. May react with ammonium salts resulting in evolution of ammonia gas.

SECTION 11: TOXICOLOGICAL INFORMATION**11.1 Information on toxicological effects**

Acute toxicity - Ingestion	May be harmful if swallowed. LD50 (rat) : 3400mg/kg
Acute toxicity - Skin Contact	Not classified.
Acute toxicity - Inhalation	Not classified.
Skin corrosion/irritation	Causes skin irritation. Sodium silicates can be irritating to corrosive to the skin of rabbits, depending on their molar ratio and concentration. Irrespective of the counterion (Na ⁺ or K ⁺), silicates were found to be corrosive at molar ratios up to 1.6 and concentrations >50%. At molar ratios >1.6, silicates are irritating to the skin, while molar ratios >3.2 and concentrations <40% did not lead to irritative effects.
Serious eye damage/irritation	Causes serious eye irritation.
Skin sensitization data	It is not a skin sensitizer.
Respiratory sensitization data	Not classified.
Germ cell mutagenicity	There is no evidence of mutagenic potential.
Carcinogenicity	No evidence of carcinogenicity.
Reproductive toxicity	No evidence of reproductive effects.
Lactation	Not classified.
STOT - single exposure	Not classified.
STOT - repeated exposure	Not classified.
Aspiration hazard	Not classified.

11.2 Other information

Not known.

SECTION 12: ECOLOGICAL INFORMATION**12.1 Toxicity**

Toxicity - Aquatic invertebrates	Low toxicity to invertebrates.
Toxicity - Fish	Low toxicity to fish.
Toxicity - Algae	Low toxicity to algae.
Toxicity - Sediment Compartment	Not classified.
Toxicity - Terrestrial Compartment	Not classified.

12.2 Persistence and degradability

The product is unlikely to persist in the environment.

12.3 Bioaccumulative potential

The product has low potential for bioaccumulation.

**Sodium Silicate Solution, MR 2.6-3.2****12.4 Mobility in soil**

Soluble in water. The product is predicted to have high mobility in soil.

12.5 Other adverse effects

Liquid is alkaline and may increase the pH. High pH can be harmful to aquatic life.

SECTION 13: DISPOSAL CONSIDERATIONS**13.1 Waste treatment methods**

Normally suitable for disposal at approved land waste site after dilution or neutralisation.

13.2 Additional Information

Disposal should be in accordance with local, state or national legislation.

SECTION 14: TRANSPORT INFORMATION

Not classified as hazardous for transport.

14.1 UN number

Not applicable

14.2 UN proper shipping name

Not applicable

14.3 Transport hazard class(es)

Not applicable

14.4 Packing group

Not applicable

14.5 Environmental hazards

Not classified as a Marine Pollutant.

14.6 Special precautions for user

Not known

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not known

SECTION 15: REGULATORY INFORMATION**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**

AICS (Australian Inventory of Chemical Substances) 1344-09-8

IMAP (Inventory Multi-Tiered Assessment and Prioritisation) – Stage One chemicals 1344-09-8

List of PECs (Priority Existing Chemicals) Not listed

NPI (National Pollutant Inventory) Not listed

Schedule 10 of WHS Regulations – Not listed

Prohibited carcinogens, restricted carcinogens and restricted hazardous chemicals

Australian Dangerous Goods Code Not applicable

Aust. Pesticides & Veterinary Medicine Authority Chemicals Not applicable

Therapeutic Goods Administration Scheduled poison S5

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Food Standards Australian & New Zealand	Not applicable
Chemicals Weapons Act	Not applicable
Ozone Depleting Substance Act	Not applicable

SECTION 16: OTHER INFORMATION

The following sections contain revisions or new statements: 1-16

LEGEND

Hazard Pictogram(s)



GHS07

Hazard classification

Skin Irrit. 2 : Skin corrosion/irritation, Category 2
Eye Irrit. 2A : Serious eye damage/irritation, Category 2A

Hazard Statement(s)

H315: Causes skin irritation.
H319: Causes serious eye irritation.

Precautionary Statement(s)

P264: Wash hands and exposed skin thoroughly after handling.
P280: Wear protective gloves/protective clothing/eye protection/face protection.
P302+P352: IF ON SKIN: Wash with plenty of water.
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P332+P313: If skin irritation occurs: Get medical advice/attention.
P337+P313: If eye irritation persists: Get medical advice/attention.
P362+P364: Take off contaminated clothing and wash it before reuse.

Acronyms

ATE : Acute Toxicity Estimate
CAS : Chemical Abstracts Service
LTEL : Long term exposure limit
STEL : Short term exposure limit
STOT : Specific Target Organ Toxicity

Key literature references and sources for data used to compile the SDS In accordance to GHS as implemented under WHS Regulations

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